Ofqual QAN	**600/0406/X**
Quartz ID	**1156, 1157**
Qualification	**Level 4 Diploma in Computing – Level 4 Diploma in Computing with Business Management**

Assessment	**Computer Network**
Assessment Type	**Examination**
Date	**Wednesday 5th March 2025**
Time	**9.00am – 12.00pm**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

© NCC Education Limited 2025

Answer ALL questions

Marks

Question 1

a) Explain in detail the purpose of the Data Link Layer in the OSI model **and** discuss its relationship to other layers. **5**

Mark Scheme: Any 5 from:

- ***The main purpose of the data link layer is to provide reliable data transfer across a physical link (1 mark).***
- ***The Data Link Layer is primarily responsible for dealing with data rates to prevent a slow receiver from being overwhelmed by a fast transmitter (1 mark).***
- ***It frames data into data frames (1 mark),***
- ***It manages access to the shared medium (1 mark),***
- ***It addresses devices using MAC addresses (1 mark),***
- ***It is at layer 2 of the OSI model (1 mark),***
- ***Adjacent layers Physical and Network (1 mark).***

b) Name ONE (1) routed protocol **and** explain its function. **5**

Mark Scheme: Any 5 from:

- ***IPv4 is a routed protocol (1 mark),***
- ***Its function is to provide a unique address to devices on a network (1 mark),***
- ***Enables the routing of data packets between different networks (1 mark),***
- ***It comprises of a network address and host address (1 mark),***
- ***It responsible for delivering packets of data from the source host to the destination host based on their IP addresses (1 mark),***
- ***It handles addressing, packet forwarding, and routing, making it a fundamental protocol for internet communication (1 mark).***

Accept other routable protocols

Total 10 marks

© NCC Education Limited 2025

Question 2

a) Explain in detail how a packet filtering firewall works. **4**

Mark Scheme:

- ***It examines the header information of data packets (1 mark),***
- ***It compares the packet's header information against a set of rules defined by the network administrator (1 mark),***
- ***These rules specify which packets are allowed or denied based IP addresses, port numbers, and protocols (1 mark),***
- ***Based on the rule matching process, the firewall decides whether to allow or block the packet (1 mark).***

b) At which OSI layer does a packet filtering firewall work? **1**

- ***A packet filtering firewall operates at the network layer of the OSI model (1 mark).***

c) Explain in detail the purpose of network security policies **and** outline TWO (2) examples. **5**

Mark Scheme:

Purpose - Any 3 from:

- ***Network security policies define the rules and guidelines for network use (1 mark),***
- ***It aims to protect the network from security threats (1 mark),***
- ***It ensures confidentiality, integrity, and availability of data and resources within a network (1 mark),***
- ***It manages user access to network resources, ensuring that only authorised users have access (1 mark).***

Example – Any 2 from:

- ***Use policy (1 mark),***
- ***Password policy (1 mark),***
- ***Incident response policy (1 mark)***

Total 10 marks

© NCC Education Limited 2025

Question 3

List **and** explain FIVE (5) security threats to a network. **10**

Mark Scheme:
Award 1 mark for a correct threat and an additional mark for elaboration: 2 for each threat.

- ***Malware: (1 mark)***
- ***Includes viruses, worms, Trojans, ransomware, and spyware. Viruses attach themselves to legitimate programs and spread when the program is executed. Worms can self-replicate and spread across networks without the need for a host program. Trojans disguise themselves as useful software but carry out malicious activities once installed. Ransomware encrypts organisation's data and demands a ransom for the decryption key. Spyware monitors a user's activities, collecting information like passwords, credit card numbers. (1 mark for any valid point)***

- ***Phishing: (1 mark)***
- ***Phishing attacks involve tricking users into revealing sensitive information, such as usernames, passwords, and credit card details. Attackers typically send emails that appear to be from legitimate sources, like banks or popular online services. These emails often contain links to fake websites that mimic the real ones, where users are then prompted to enter their credentials. (1 mark)***

- ***Denial of Service (DoS) / (DDoS) attacks: (1 mark)***
- ***An attacker floods a target server, network, or service with traffic, overwhelming it and making it unavailable to legitimate users often using a botnet (1 mark)***

- ***Man-in-the-Middle (MitM) attacks: (1 mark)***
- ***An attacker intercepts communication between two parties, such as a user and a server. The attacker can then eavesdrop on the conversation, modify the data being transmitted (1 mark)***

- ***Insider Threats: (1 mark)***
- ***Come from within an organisation, such as employees, contractors, or partners. An insider with malicious intent could steal sensitive data, sabotage systems, or misappropriate company resources. (1 mark)***

Total 10 marks

© NCC Education Limited 2025

Question 4

a) Explain the key characteristics of a client-server network **and** discuss its advantages over a peer-to-peer network. **4**

Mark Scheme: Any 4 from:

- ***Centralised Resources: A client-server network offers centralised resources such as files, applications, and services are centralised on a server (1 mark),***
- ***Security: A client-server network offers centralised control allows for better implementation of security policies. Access to resources can be managed and monitored more effectively. (1 mark),***
- ***Scalability: Client-server networks are highly scalable. Additional clients and servers can be added to the network without significant changes to the existing infrastructure. (1 mark),***
- ***Reliability: Servers are typically more robust and reliable than individual client machines. They often have redundancy and backup systems in place to ensure continuous availability. (1 mark),***
- ***Award a mark for a suitable diagram with explanation. (1 mark).***
- ***In contrast, peer-to-peer networks distribute responsibilities among all devices, which can lead to challenges in management, security, and scalability. (1 mark)***

b) Explain in detail the difference between a physical network topology **and** a logical network topology. **6**

Mark Scheme: Any 3 bullet points with 2 marks for each bullet point:

- ***Physical topology refers to the physical layout of devices and cables in a network (1 mark), while logical topology is about the data paths in the network, focusing on how data flows within the network (1 mark).***
- ***Physical topology focuses on the physical arrangement and connections of network devices (1 mark), while logical topology focuses on the logical flow of data and the virtual connections between devices (1 mark).***
- ***Physical topology involves the actual installation and setup of network hardware (1 mark) while logical topology involves configuring network protocols, addressing schemes, and virtual connections (1 mark).***
- ***Physical topology less flexible, as changes require physical modifications to the network infrastructure (1 mark), while logical topology more flexible, as changes can often be made through software configurations without altering the physical setup (1 mark).***

Total 10 marks

© NCC Education Limited 2025

Question 5

a) Explain in detail the function of a **switch** in a network. You should clearly identify the OSI layer at which it operates. **5**

Mark Scheme:

- ***A switch operates at the data link layer (1 mark).***
- ***A network switch is a device that connects multiple devices within a local area network (LAN) (1 mark),***
- ***It receives data packets from connected devices and forwards them to the appropriate destination device within the network (1 mark),***
- ***It uses the MAC (Media Access Control) addresses of devices to determine the correct port to send the data to (1 mark),***
- ***It can prioritise certain types of traffic to ensure that critical applications receive the necessary bandwidth (1 mark).***

b) Explain in detail the function of a **router** in a network. You should clearly identify the OSI layer at which it operates. **5**

Mark Scheme:

- ***A router works at the network layer (1 mark).***
- ***It connects two or more networks together and directs data packets between them (1 mark)***
- ***Its main function is to choose the route for packets on different networks (1 mark),***
- ***Routers maintain a routing table that is used to make forwarding decisions, ensuring that data packets take the most efficient path to their destination. (1 mark)***
- ***Routers can perform Network Address Translation (NAT), which allows multiple devices on a private network to share a single public IP address. (1 mark)***

Total 10 marks

© NCC Education Limited 2025

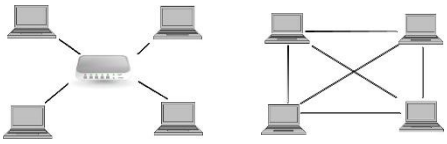
Question 6

Star and mesh are TWO (2) common network topologies. Compare **and** contrast these topologies by completing the following:

a) Draw a diagram to illustrate each of them. **4**

Mark Scheme: 2 marks for each correct diagram:

- ***Star should include label for hub/switch, Mesh should show multiple interconnections***



b) Describe their relative advantages **and** disadvantages. **4**

Star Topology:

- ***Advantages: More reliable as the failure of one cable or device does not impact the entire network. It is also easy to troubleshoot and maintain (1 mark).***
- ***Disadvantages: Requires more cabling, which increases the cost. The central hub or switch is a potential single point of failure (1 mark).***

Mesh Topology:

- ***Advantages: High reliability and fault tolerance due to multiple redundant paths between nodes. It can handle a large amount of traffic efficiently (1 mark).***
- ***Disadvantages: Complex to set up and manage, and requires a large number of connections, which can be costly in terms of hardware and configuration (1 mark).***

c) Identify suitable applications for each topology. **2**

Star Topology:

- ***Application: Suitable for medium - sized offices where reliability and ease of management are important, such as a corporate office with multiple departments (1 mark).***

Mesh Topology:

- ***Application: Used in critical environments like military networks (1 mark)***
- ***or large - scale data centres (1 mark) where high reliability and continuous operation are essential.***

1 mark for Star Topology application and 1 mark for Mesh Topology application.

Total 10 marks

© NCC Education Limited 2025

Question 7

Explain the process of installing **and** configuring a firewall on an internet-connected system. **10**

Your answer should include the SIX (6) key steps and considerations

Mark Scheme: 1 mark for identifying a suitable step + 1 for elaboration to a maximum of 10 marks
Eg:

- ***Planning: (1 mark)***
  - ***First, determine the security requirements of the network, including what traffic to allow or block (1 mark).***
  - ***This involves identifying potential threats and the level of security needed (1 mark).***
- ***Selection: (1 mark)***
  - ***Choose a firewall solution, either a software - based firewall for individual computers or a hardware - based firewall for network - wide protection. (1 mark)***
  - ***Consider factors like ease of use, features, and cost (1 mark).***
- ***Installation: (1 mark)***
  - ***For a software - based firewall, install the software on the relevant devices (1 mark).***
  - ***For a hardware - based firewall, physically install the device in the network, connecting it to the appropriate network segments (1 mark).***
- ***Configuration: (1 mark)***
  - ***Set up rules to control traffic (1 mark).***
  - ***This can include blocking specific IP addresses, ports, or protocols. (1 mark)***
  - ***For example, block incoming traffic from known malicious IPs or block ports commonly used for attacks (1 mark).***
- ***Testing: (1 mark)***
  - ***Thoroughly test the firewall to ensure that it allows legitimate traffic and blocks unauthorised traffic (1 mark).***
  - ***Check for any interference with existing software applications (1 mark).***
- ***Maintenance: (1 mark)***
  - ***Regularly update the firewall's rules, software, and signatures to adapt to new threats and network changes (1 mark).***

Total 10 marks

© NCC Education Limited 2025

Question 8

a) Describe the FOUR (4) components of a VoIP system. **4**

Mark Scheme:

- ***Servers: Manage calls and process connections. They handle tasks like call routing and authentication (1 mark).***
- ***Endpoint devices: Convert to and from analogue voice systems, such as telephone handsets, computer speakers/microphones. They are the devices through which users interact with the VoIP system (1 mark).***
- ***Gateways: Connect IP and non - IP devices, allowing VoIP networks to communicate with traditional telephone networks (1 mark).***
- ***IP networks: Provide the infrastructure for transmitting voice data as packets (1 mark).***

b) Explain how these components work together. **2**

Mark Scheme:

Endpoint devices capture analogue voice signals and convert them into digital packets. (1 mark) These packets are sent over the IP network. Servers manage the call - related processes, and gateways enable communication with non - IP devices if needed (1 mark).

c) Discuss FOUR (4) security issues associated with VoIP. **4**

Mark Scheme:

- ***Hacking: VoIP systems can be hacked to access sensitive information or disrupt services (1 mark).***
- ***Eavesdropping: As voice data is transmitted over the network, it can be intercepted by unauthorised parties (1 mark).***
- ***Denial of Service (DoS) attacks: Flooding the VoIP network with requests can disrupt normal call services (1 mark).***
- ***Lack of encryption: If not properly encrypted, voice data can be easily accessed and understood by attackers (1 mark).***

Total 10 marks

© NCC Education Limited 2025

Question 9

Network Security is often defined in terms of THREE (3) key objectives expressed by the acronym CIA.

a) State **and** explain each of these THREE (3) objectives. **6**

Mark Scheme:

- ***Confidentiality: (1 mark)***
  - ***Ensures that only authorised users have access to network services and data. (1 mark)***
- ***Integrity: (1 mark)***
  - ***Guarantees that the data sent and received is the same, without any modification during transmission. (1 mark)***
- ***Availability: (1 mark)***
  - ***Ensures that the network is available for users when needed. (1 mark)***

b) Give examples of how technology can help achieve these objectives. **4**

© NCC Education Limited 2025

Mark Scheme:
Max 4 marks - 2 marks for any one of CIA:

- ***Confidentiality***
  - ***Encryption (1 mark): Encryption technologies such as AES (Advanced Encryption Standard) is widely used to protect sensitive information in transit and at rest, preventing unauthorised access (1 mark).***
  - ***Access Control Systems (1 mark): Technologies like Multi-Factor Authentication (MFA) and Role-Based Access Control (RBAC) ensure that only authorised individuals can access certain data and systems, maintaining confidentiality (1 mark).***
- ***Integrity***
  - ***Hashing Algorithms (1 mark): Hashing algorithms like SHA-256 create unique digital fingerprints of data. These hashes can be used to verify that data has not been altered, ensuring its integrity (1 mark).***
  - ***Digital Signatures (1 mark): Digital signatures use cryptographic techniques to verify the authenticity and integrity of a message, software, or digital document. They ensure that the data has not been tampered with and confirm the identity of the sender (1 mark).***
- ***Availability***
  - ***Redundant Systems (1 mark): Technologies such as RAID (Redundant Array of Independent Disks) and failover clustering ensure that systems remain operational even if one component fails. This enhances the availability of data and services. (1 mark).***
  - ***Load Balancers (1 mark): Load balancers distribute network or application traffic across multiple servers to ensure no single server becomes overwhelmed. This helps maintain the availability and performance of services (1 mark).***

Total 10 marks

© NCC Education Limited 2025

Question 10

a) Define what is meant by a VPN. **3**

Mark Scheme: Any 3 from:

- ***A VPN creates a virtual private (secure) connection (1 mark)***
- ***Over a shared (untrusted) network, (1 mark) typically the Internet.***
- ***From a user's perspective, it appears as a private dedicated connection. (1 mark)***
- ***It enables organisations and individuals to communicate privately using the public Internet infrastructure. (1 mark)***

b) Explain clearly **and** succinctly how a VPN works. **5**

Mark Scheme: Any 5 from:

- ***VPNs use encryption to protect data. (1 mark)***
- ***Public key encryption and digital signatures are commonly used for authentication and encryption. (1 mark)***
- ***When data is sent through a VPN, it is encrypted at the source, travels through the VPN tunnel over the Internet, and is decrypted at the destination. (1 mark)***
- ***This ensures confidentiality, integrity, and authentication of the data being transmitted. (1 mark)***
- ***Additionally, VPNs can also mask the user's IP address, providing an extra layer of privacy. (1 mark)***
- ***Connections: There are connections to the Internet and the VPN itself. (1 mark)***
- ***Devices need to be connected to the Internet first to access the VPN service. (1 mark)***
- ***Firewalls: Firewalls are used to control what traffic is allowed to pass through the VPN connection, adding an extra layer of security. (1 mark)***

Any other valid points (eg site-to site VPN / Remote access VPN)

c) Identify TWO (2) protocols used in a VPN **2**

Mark Scheme:
Any Two (2) Protocols (Any 2 of the below up to 2 marks):

- ***OpenVPN:.***
- ***L2TP (Layer 2 Tunnelling Protocol):***
- ***PPTP (Point - to - Point Tunnelling Protocol):***
- ***SSL/TLS***
- ***IPsec***

Total 10 marks

End of paper

© NCC Education Limited 2025

Learning Outcomes Matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1	Yes
2	3	Yes
3	3	Yes
4	4	Yes
5	2,4	Yes
6	2	Yes
7	3	Yes
8	6	Yes
9	3	Yes
10	6	Yes

Grade descriptors					
Learning Outcome	**Fail**	**Referral**	**Pass**	**Merit**	**Distinction**
Understand network and communication protocols	None or inadequate description of network and communication protocols	Some credible but incomplete attempt to demonstrate understanding of network and communication protocols	Limited but satisfactory demonstration of understanding of network and communication protocols	Good demonstration of understanding of network and communication protocols	Excellent demonstration of understanding of network and communication protocols
Understand the principles of common network topologies and architectures	None or very basic description of common network topologies and architectures	A very limited and basic description of common network topologies and architectures	Limited and above basic description of some common network topologies and architectures	Good description of most common network topologies and architectures	Excellent and full description of all common network topologies and architectures
Understand the application of network security measures	No demonstration of application of network security measures	A basic demonstration of application of network security measures	An above basic demonstration of application of network security measures	A good demonstration of application of network security measures	An excellent demonstration of application of network security measures
Be able to select and configure the hardware components of a computer network to meet the	None or trivial evidence of ability to select and configure the hardware components of a computer	Limited evidence of ability to select and configure the hardware components of a computer	Basic evidence of ability to select and configure the hardware components of a computer network	Good evidence of ability to select and configure the hardware components of a computer	Excellent evidence of ability to select and configure the hardware components of a computer

© NCC Education Limited 2025

requirements of a precise specification.	network to meet the requirements	network to meet the requirements	to meet the requirements	network to meet the requirements	network to meet the requirements
Be able to design and install network and server operating systems to meet the requirements of a precise specification	No or very limited evidence of ability to design and install network and server operating systems	Limited and very basic evidence of ability to design and install network and server operating systems	Basic but limited evidence of ability to design and install network and server operating systems	Good and competent evidence of ability to design and install network and server operating systems	Excellent and complete evidence of ability to design and install network and server operating systems
Be able to define and configure internet telephony and communication systems	No or insufficient evidence of ability to define and configure internet telephony and communication systems	Limited and very basic evidence of ability to define and configure internet telephony and communication systems	Basic and partly satisfactory evidence of ability to define and configure internet telephony and communication systems	Good evidence of ability to define and configure internet telephony and communication systems	Excellent evidence of ability to define and configure internet telephony and communication systems

© NCC Education Limited 2025